

CS2601 Linear and Convex Optimization

Homework 3

Due: 2024.11.22

For this assignment, you should submit a **single** pdf file as well as your source code (.py or .ipynb files). **The pdf file should include all necessary figures, the outputs of your Python code, and your answers to the questions.** Do NOT submit your figures in separate files. Your answers in any of the .py or .ipynb files will NOT be graded.

For problem 5 and 6, you will implement gradient descent with constant step size. First complete the function `gd_const_ss` in `gd.py`.

1. Is the following set convex? Explain your argument.

(a).

$$S = \{x \in \mathbb{R}^2 : x > \mathbf{0}, x_1 \log x_1 + x_2 \log x_2 \leq 2\}.$$

(b).

$$S = \{\mathbf{x} = (x_1, x_2) \in \mathbb{R}^2 : \max\{\|\mathbf{A}\mathbf{x} + \mathbf{b}\|^3, \log(1 + e^{5x_1 + 3x_2})\} \leq 2\}.$$

2. Let $\bar{B} = \{\mathbf{x} \in \mathbb{R}^n : \|\mathbf{x}\| \leq 1\}$ be the closed unit 2-norm ball. Recall the projection of a point $\mathbf{x}_0$ onto $\bar{B}$ is the solution to the following problem,

$$\min_{\mathbf{x} \in \bar{B}} f(\mathbf{x}) = \frac{1}{2} \|\mathbf{x} - \mathbf{x}_0\|^2$$

Use the first-order optimality condition to show that the projection of $\mathbf{x}_0 \notin \bar{B}$ is $\hat{\mathbf{x}}_0 = \frac{\mathbf{x}_0}{\|\mathbf{x}_0\|}$.

3. Determine if the following optimization problems are convex optimization problems.

(a).

$$\begin{aligned} \min_{x_1, x_2} \quad & x_1^2 - 2x_1x_2 + x_2^2 + x_1 + x_2 \\ \text{s.t.} \quad & x_1 e^{-(x_1 + x_2)} \leq 0 \\ & x_1^2 - 3x_2 = 0 \end{aligned}$$

(b).

$$\begin{aligned} \min_{x_1, x_2} \quad & x_1^2 + x_2^4 \\ \text{s.t.} \quad & (x_1 - x_2)^2 + 4x_1x_2 + e^{x_1+x_2} \leq 0 \\ & x_1^2 - 2x_1x_2 + x_2^2 + x_1 + x_2 \leq 0 \\ & 6x_1 - 7x_2 = 0 \end{aligned}$$

4. Consider the optimization problem

$$\begin{aligned} \min_{x_1, x_2} \quad & f(x_1, x_2) \\ \text{s.t.} \quad & 4x_1 + x_2 \geq 1 \\ & 2x_1 + 3x_2 \geq 1 \\ & x_1 \geq 0, x_2 \geq 0 \end{aligned}$$

Sketch the feasible set. Find the set of optimal solutions and the optimal value graphically.

(a). $f(x_1, x_2) = x_1 + x_2$

(b). $f(x_1, x_2) = -x_1 - x_2$

5. Consider the following quadratic program,

$$\min_{\mathbf{x} \in \mathbb{R}^2} f(\mathbf{x}) = \frac{1}{2} \mathbf{x}^T \mathbf{Q} \mathbf{x} \tag{1}$$

where

$$\mathbf{Q} = \begin{pmatrix} \gamma & 0 \\ 0 & 1 \end{pmatrix}.$$

with $\gamma > 0$. In terms of coordinates,

$$f(x_1, x_2) = \frac{\gamma}{2} x_1^2 + \frac{1}{2} x_2^2.$$

We know the optimal solution is $\mathbf{x}^* = \mathbf{0}$ and $f^* = f(\mathbf{x}^*) = 0$.

(a). What is the smallest L such that f is L -smooth?

(b). Suppose $\gamma = 0.1$. Complete **p5.py** and run the gradient descent algorithm you implemented with $\mathbf{x}_0 = (1, 1)$ and step sizes 2.2, 1, 0.5, 0.01. (Note: for step size 2.2, you may want to limit the maximum number of iterations to a small number, say 10.) Does it converge in each case? When it does converge, how many iterations does it take? In each case, plot the 2D trajectory of the sequence $\mathbf{x}_k$ and the function values $f(\mathbf{x}_k)$. You can use the function provided in **utils.py** or write your own code. Note that $f(\mathbf{x}_k)$ is also the suboptimality gap since $f(\mathbf{x}^*) = 0$.

(c). For $\gamma = 1, 0.1, 0.01, 0.001$, run gradient descent with step size 1. How many iterations do you need in each case? How does the number of iterations change as γ decreases? You don't need to show the plots.

6. Linear regression. Consider the least squares problem as follows,

$$\min_{\mathbf{w}} \|\mathbf{X}\mathbf{w} - \mathbf{y}\|_2^2$$

where

$$\mathbf{X} = \begin{pmatrix} 2 & 0 \\ 0 & 1 \\ 0 & 0 \end{pmatrix}, \quad \mathbf{y} = \begin{pmatrix} 4 \\ 1 \\ 2 \end{pmatrix}.$$

- (a). Solve the least squares problem in closed form.
- (b). Use your implementation of gradient descent to solve this least squares problem. You may reuse the codes for Problem 4 by rewriting `f` and `fp` (you don't have to plot the traces as in Problem 4). Use any stepsize you find appropriate and show your results. Compare your solution with the closed-form solution in 5(a). Do they agree with each other?